

Lab 2: Verification of the Superposition Principle

ESC 221 Basic Principles of Electrical Engineering

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I. Introduction, Experimental Method and Hypotheses

A. Introduction

This experiment aims to verify the superposition principle in linear circuits. The superposition principle states that in a linear circuit with multiple independent sources, the voltage or current at any point in the circuit can be found by summing the contributions from each independent source acting individually while all other sources are turned off.

B. Experimental Method

In identical resistor networks as shown in Figures 1, 2, and 3, a multimeter will be used to measure the voltages at nodes A and B for voltage sources V_1 and V_2 individually and then simultaneously. This approach allows for the determination of each source's contribution to the network, which will then be summed to verify the superposition principle for the nodal voltages at A and B in the circuit with both sources active.

C. Hypotheses

It is predicted that the measured voltages at nodes A and B in circuit C (Fig. 3) will be equal to the sum of the voltages at nodes A and B in circuits A and B (Figs. 1 and 2), thus confirming the superposition principle for this linear circuit network.

$$V_{A_{\text{circuit C}}} = V_{A_{\text{circuit A}}} + V_{A_{\text{circuit B}}}$$

$$V_{B_{\text{circuit C}}} = V_{B_{\text{circuit A}}} + V_{B_{\text{circuit B}}}$$

Based on the nodal analysis calculations shown in the appendix, the predicted nodal voltages for each circuit configuration are as follows:

- Circuit A (V_1 only): $V_A = 2.2350 \text{ V}$, $V_B = 1.0469 \text{ V}$
- Circuit B (V_2 only): $V_A = 0.4444 \text{ V}$, $V_B = 2.4002 \text{ V}$
- Circuit C (V_1 and V_2): $V_A = 2.6803 \text{ V}$, $V_B = 3.4476 \text{ V}$

II. Equipment List

1. **Digital Multimeter**

Manufacturer: AstroAI

Model: AM33B

2. **DC Power Supply**

Manufacturer: GW Instek

Model: SPD-3606

III. Wiring / Schematic Diagram / Procedure

A. Procedure

1. Constructing Circuit A

1. A circuit was constructed as illustrated in Fig. 1 using a DC power supply ($V_1 = 5\text{ V}$) and resistors ($R_1 = 1k\ \Omega$, $R_2 = 1k\ \Omega$, $R_3 = 2.2k\ \Omega$, $R_4 = 3.3k\ \Omega$, and $R_5 = 4.7k\ \Omega$).
2. Resistor R_1 was connected between the positive terminal of V_1 and node A.
3. Resistor R_2 was connected between node A and ground.
4. Resistor R_3 was connected between node A and node B.
5. Resistor R_4 was connected between node B and ground.
6. Resistor R_5 was connected between node B and ground.
7. The negative terminal of V_1 was connected to ground.

2. Constructing Circuit B

1. A circuit was constructed as illustrated in Fig. 2 using a DC power supply ($V_2 = 10\text{ V}$) and resistors ($R_1 = 1k\ \Omega$, $R_2 = 1k\ \Omega$, $R_3 = 2.2k\ \Omega$, $R_4 = 3.3k\ \Omega$, and $R_5 = 4.7k\ \Omega$).
2. Resistor R_1 was connected between node A and ground.
3. Resistor R_2 was connected between node A and ground.
4. Resistor R_3 was connected between node A and node B.
5. Resistor R_4 was connected between node B and ground.
6. Resistor R_5 was connected between node B and the negative terminal of V_2 .
7. The negative terminal of V_2 was connected to ground.

3. Constructing Circuit C

1. A circuit was constructed as illustrated in Fig. 3 using both DC power supplies ($V_1 = 5\text{ V}$ and $V_2 = 10\text{ V}$) and resistors ($R_1 = 1k\ \Omega$, $R_2 = 1k\ \Omega$, $R_3 = 2.2k\ \Omega$, $R_4 = 3.3k\ \Omega$, and $R_5 = 4.7k\ \Omega$).
2. Resistor R_1 was connected between the positive terminal of V_1 and node A.
3. Resistor R_2 was connected between node A and ground.
4. Resistor R_3 was connected between node A and node B.
5. Resistor R_4 was connected between node B and ground.

6. Resistor R_5 was connected between node B and the negative terminal of V_2 .
 7. The negative terminals of V_1 and V_2 were both connected to ground.
4. Measurements and Data Collection
1. For each circuit configuration (A, B, and C), the voltages at nodes A and B were measured using a multimeter with the positive lead connected to nodes A and B, respectively, and the negative lead connected to ground.

B. Schematic Diagram

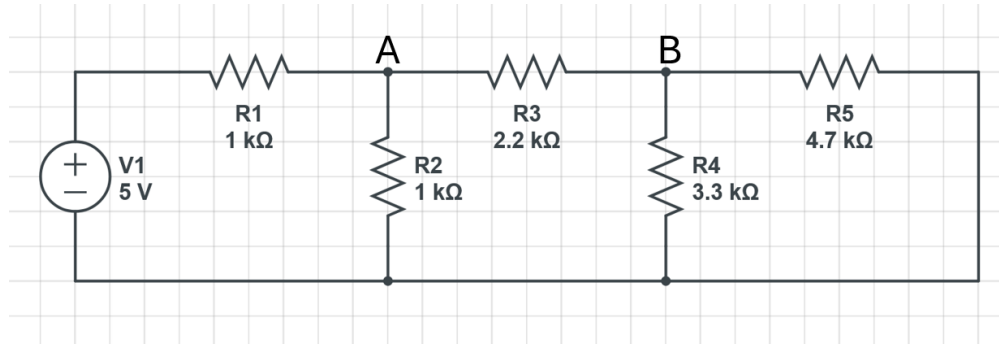


Figure 1: Circuit A: A DC circuit network consisting of a fixed voltage source V_1 and resistors R_1 , R_2 , R_3 , R_4 , and R_5 with labeled nodes A and B for measurement points.

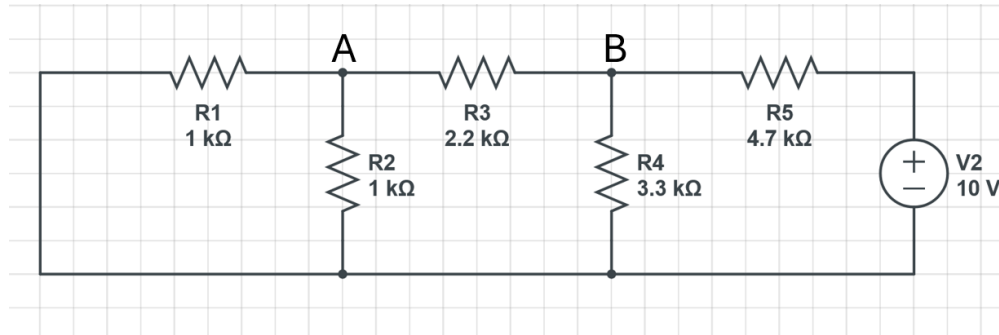


Figure 2: Circuit B: A DC circuit network consisting of a fixed voltage source V_2 and resistors R_1 , R_2 , R_3 , R_4 , and R_5 with labeled nodes A and B for measurement points.

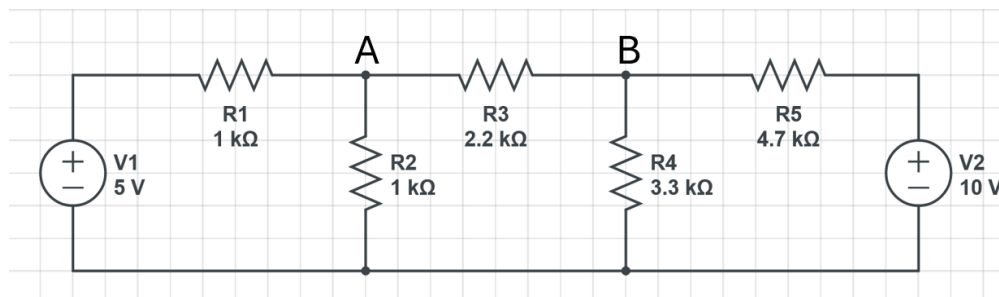


Figure 3: Circuit C: A DC circuit network consisting of both voltage sources V_1 and V_2 and resistors R_1 , R_2 , R_3 , R_4 , and R_5 with labeled nodes A and B for measurement points.

IV. Results

Table 1: Comparison of Calculated and Measured Nodal Voltages

Nodal Voltage (V)	Circuit A ($V_1=5\text{V}$)	Circuit B ($V_2=10\text{V}$)	Circuit C (V_1 & V_2)
<i>Calculated Values</i>			
V_A	2.2350	0.4444	2.6803
V_B	1.0469	2.4002	3.4476
<i>Measured Values</i>			
V_A	2.29	0.43	2.75
V_B	1.04	2.34	3.38

V. Discussion, Analysis and Concluding Statements

A. Verification of Superposition with Calculated Values

Comparing the calculated nodal voltages for circuits A, B, and C using nodal analysis, the sum of the voltages at nodes A and B for circuits A and B closely matches the calculated voltages for circuit C (some difference due to rounding), confirming the superposition principle.

$$\begin{aligned}V_{A_{\text{circuit C}}} &= V_{A_{\text{circuit A}}} + V_{A_{\text{circuit B}}} \\ \implies 2.6803 \text{ V} &\approx 2.2350 \text{ V} + 0.4444 \text{ V} = 2.6794 \text{ V} \\ V_{B_{\text{circuit C}}} &= V_{B_{\text{circuit A}}} + V_{B_{\text{circuit B}}} \\ \implies 3.4476 \text{ V} &\approx 1.0469 \text{ V} + 2.4002 \text{ V} = 3.4471 \text{ V}\end{aligned}$$

B. Verification of Superposition with Measured Values

Comparing the measured nodal voltages for circuits A, B, and C in Table 1, the sum of the voltages at nodes A and B for circuits A and B closely matches the measured voltages for circuit C, confirming the superposition principle in practice.

$$\begin{aligned}V_{A_{\text{circuit C}}} &= V_{A_{\text{circuit A}}} + V_{A_{\text{circuit B}}} \\ \implies 2.75 \text{ V} &\approx 2.29 \text{ V} + 0.43 \text{ V} = 2.72 \text{ V} \\ V_{B_{\text{circuit C}}} &= V_{B_{\text{circuit A}}} + V_{B_{\text{circuit B}}} \\ \implies 3.38 \text{ V} &\approx 1.04 \text{ V} + 2.34 \text{ V} = 3.38 \text{ V}\end{aligned}$$

C. Conclusion

This experiment successfully verified the superposition principle in linear circuits. Both the calculated and measured nodal voltages at nodes A and B for circuit C closely matched the sum of the voltages from circuits A and B, confirming that the contributions from each independent source can be linearly added to determine the overall response in a linear circuit network. Small discrepancies in the verification of superposition with measured values can be attributed to factors such as resistor tolerances, fluctuations in the voltage readings, and measurement uncertainties. Overall, the results support the validity of the superposition principle.

VI. Appendix

Calculations for Predicated Nodal Voltages

Using nodal analysis, the predicted voltages at nodes A and B for circuits A, B, and C can be calculated as follows :

A. Circuit A

Using $V_1 = 5\text{ V}$, $R_1 = 1k\ \Omega$, $R_2 = 1k\ \Omega$, $R_3 = 2.2k\ \Omega$, $R_4 = 3.3k\ \Omega$, and $R_5 = 4.7k\ \Omega$.

1. Node A KCL Equation:

$$\frac{V_1 - V_A}{R_1} = \frac{V_A - V_B}{R_3} + \frac{V_A}{R_2} \implies 0.005 = 0.00245V_A - 0.0004545V_B$$

2. Node B KCL Equation:

$$\frac{V_A - V_B}{R_3} = \frac{V_B}{R_4} + \frac{V_B}{R_5} \implies 0 = 0.0004545V_A - 0.0009703V_B$$

Solving the system of equations yields:

$$V_A = 2.2350\text{ V}$$

$$V_B = 1.0469\text{ V}$$

B. Circuit B

Using $V_2 = 10\text{ V}$, $R_1 = 1k\ \Omega$, $R_2 = 1k\ \Omega$, $R_3 = 2.2k\ \Omega$, $R_4 = 3.3k\ \Omega$, and $R_5 = 4.7k\ \Omega$.

1. Node B KCL Equation:

$$\frac{V_2 - V_B}{R_5} = \frac{V_B}{R_4} + \frac{V_B - V_A}{R_3} \implies 0.002127 = -0.0004545V_A + 0.0009703V_B$$

2. Node A KCL Equation:

$$\frac{V_B - V_A}{R_3} = \frac{V_A}{R_2} + \frac{V_A}{R_1} \implies 0 = 0.0024545V_A - 0.0004545V_B$$

Solving the system of equations yields:

$$V_A = 0.4444\text{ V}$$

$$V_B = 2.4002\text{ V}$$

C. Circuit C

Using $V_1 = 5\text{ V}$, $V_2 = 10\text{ V}$, $R_1 = 1k\ \Omega$, $R_2 = 1k\ \Omega$, $R_3 = 2.2k\ \Omega$, $R_4 = 3.3k\ \Omega$, and $R_5 = 4.7k\ \Omega$.

1. Node A KCL Equation:

$$\frac{V_1 - V_A}{R_1} = \frac{V_A - V_B}{R_3} + \frac{V_A}{R_2} \implies 0.005 = 0.00245V_A - 0.0004545V_B$$

2. Node B KCL Equation:

$$\frac{V_A - V_B}{R_3} = \frac{V_B}{R_4} + \frac{V_B - V_2}{R_5} \implies -0.002127 = 0.0004545V_A - 0.0009703V_B$$

Solving the system of equations yields:

$$V_A = 2.6803 \text{ V}$$

$$V_B = 3.4476 \text{ V}$$